APEX HORIZON NATIONAL BANK

Online Banking Simulator & Administrative Supervisory System — Technical Architecture Report

Version: 3.4.2-SECURE | Host: localhost:3000 | Verified Build: Production Ready

1. Executive Summary

The **Apex Horizon Online Banking Simulator** is an authentic, full-featured retail banking web application crafted to replicate modern Tier-1 financial institutions (e.g., JPMorgan Chase, Citibank, and Bank of America).

The platform provides a complete banking lifecycle across two distinct user roles:

- **Customer Self-Service:** Online account opening/onboarding, multi-account ledger balances (Premier Checking and High-Yield Savings), Fedwire/ACH funds transfers with real-time balance settlement, transaction statements with search/filter/CSV export, and customer service requests (Credit Cards and Chequebooks).
- **Bank Administration / Officer Console:** Central bank liquidity & vault oversight, patron user directory management with compliance account masking, and an interactive approval queue with Approve, Decline, and Delete capabilities.
- **Security & Compliance:** 256-Bit TLS simulation, dynamic 5-minute inactivity session expiration with countdown timer, anti-fraud notifications, GLBA data masking, and FDIC regulatory disclosures.

2. Technology Stack & Architectural Design

2.1 Front-End Layer

- **HTML5:** Semantic architecture using native landmarks (header, nav, aside, main, section, footer) with ARIA compliance.
- **Tailwind CSS & Bespoke Tokens:** Curated executive palette featuring Bank Navy (#0b192c, #060f1c), Steel Blues (#1e4f79, #286496), and status colors (Emerald #10b981, Gold #c5a059, Crimson #ef4444).
- **Typography:** Google Fonts *Plus Jakarta Sans* for modern UI readability and *JetBrains Mono* for tabular numbers and balances.
- **Iconography:** Lucide Icons dynamic vector engine.

2.2 Single Page Application (SPA) Engine

- **Client-Side Routing:** Zero-reload dynamic DOM view manipulation switching seamlessly between Login, Customer Dashboard, and Admin Console.

- **Sub-Page Isolation:** Dedicated view isolation ensuring Customer subpages and Admin subpages never collide or leak into adjacent portals.
- **Cross-Tab Synchronizer:** Window storage event listener providing instant, bidirectional real-time synchronization between concurrent browser tabs.

2.3 Mock Backend & Data Persistence

- **Storage Engine:** Browser localStorage functioning as an atomic JSON document store under key apex_horizon_bank_v1.
- **Relational Integrity:** Automatic linkage between User IDs, debit/credit transactions, and service request filings.
- **Bilateral Clearing Simulation:** Transfers between registered patrons automatically identify the recipient account, crediting their balance while debiting the sender.

3. Core Functional Modules

3.1 Authentication & Onboarding Portal

• **Dual-Mode Access Control:** Seamless toggle tabs between Customer Portal and Bank Officer / Admin Portal with role validation.

• **Self-Service Account Creation:** Direct online enrollment creating Checking (100XXXXXXX) and High-Yield Savings (200XXXXXXX) accounts with initial deposits.

• **Credentials Assistance:** Quick Demo Fillers (jane / 1234, admin / admin), show/hide password toggle, and Forgot Password recovery guidance.

3.2 Customer Banking Experience

• **Account Overview:** Real-time balances for Premier Checking, High-Yield Growth Savings (4.85% APY), and combined net liquid portfolio with credit score indicators.

• **Funds Transfer Module (ACH & Fedwire):** Source account selector, beneficiary routing verification (9-digit ABA), purpose classification, and strict balance validation.

• **Self-Account Transfer:** Instant modal to move funds between personal Checking and Savings accounts.

• **Ledger & Statement Export:** Chronological transaction records with dynamic search, account filter, type filter (Debit/Credit), sorting, and instant CSV statement download.

• **Service Requests:** Tiered Credit Card applications and MICR chequebook leaf orders with live status tracking and cancellation capabilities.

3.3 Bank Administration Console

• **Central Vault & Liquidity Monitor:** Supervisory metrics tracking active depositors, total liquidity, pending compliance reviews, and global network transaction stream.

• **User Directory Management:** Roster of all registered customers with masked account details (•••• •••• 4891), balance breakdowns, full patron audit modals, and manual patron onboarding.

• **Approval Queue:** Complete administrative triage permitting officers to **Approve**, **Decline**, or permanently **Delete/Purge** filings.

4. Database Schema Specification (LocalStorage)

Entity	Attribute	Type	Description
User	id	String (PK)	Unique patron identifier (e.g. usr_001)
	username / password	String	Authentication credentials
	checkingAcc / savingsAcc	String	10-digit account identifiers
	checkingBalance / savingsBalance	Number	Cleared liquid account balances
Transaction	id	String (PK)	Unique transaction tracking ID (e.g. TXN-894101)
	userId / accountType	String (FK)	Owner binding and account classification
	amount / balanceAfter	Number	Monetary amount and post-settlement balance
	type / status	String	credit / debit flag and settlement state

ServiceRequest	id	String (PK)	Filing identifier (e.g. REQ-1049)
	serviceType / details	String	Card tier or Chequebook specifications
	status	String	pending | approved | rejected

5. Test Execution Matrix

Test ID	Scenario & Scope	Execution Verification Criteria	Result
TC-01	Dual Portal Switching	Switch between Customer & Admin portals; inputs & titles adapt	**PASS**
TC-02	Customer Authentication	Sign in as jane / 1234; routes directly to Customer Overview	**PASS**
TC-03	Overdraft Prevention	Attempt transfer exceeding balance; blocked with alert toast	**PASS**
TC-04	ACH / Wire Transfer	Transfer $250; balance drops, debit entry added to ledger	**PASS**
TC-05	Inter-User Clearing	Send funds to peer; credits beneficiary account in real-time	**PASS**
TC-06	Self-Account Transfer	Move $500 between Checking & Savings via modal	**PASS**
TC-07	Ledger Filtering & Search	Filter by type and query string; responsive table updates	**PASS**
TC-08	Statement CSV Export	Click export statement; downloads populated .csv file	**PASS**
TC-09	Product Applications	Submit Credit Card & Chequebook requests; shows Pending	**PASS**
TC-10	Admin Approvals	Admin clicks Approve / Decline; updates badges and tables	**PASS**
TC-11	Request Deletion	Customer cancels filing; admin deletes record via trash icon	**PASS**
TC-12	Cross-Tab Sync	Admin updates status in Tab A; Tab B reflects immediately	**PASS**
TC-13	Online Self-Registration	Open account online; creates accounts & opens dashboard	**PASS**
TC-14	Database Demo Reset	Click Reset DB; cleans state and restores factory seed data	**PASS**

6. Production Roadmap & Conclusion

The **Apex Horizon Online Banking Simulator** successfully delivers an enterprise-grade retail banking simulation meeting all functional, aesthetic, and architectural requirements.

Recommended Enterprise Extensions:

1. **Backend REST/GraphQL Microservices:** Migrate from client-side localStorage to a Node.js/Express, Python/FastAPI, or Java Spring Boot backend paired with PostgreSQL.
2. **Cryptographic Security:** Introduce salted BCrypt password hashing and JWT/OAuth2 bearer token session authorization.
3. **Third-Party Financial Integrations:** Connect Plaid or Stripe Treasury sandbox APIs for simulated external bank linking and merchant settlement.